

Multiple Choice (18 marks)

1. If $c > 0$ and $f(x) = e^x - cx$ for all real numbers x , then the minimum value of f is
 - (a) $f(c)$
 - (b) $f(e^c)$
 - (c) $f(1/c)$
 - (d) $f(\log c)$
2. If A is a subset of the real line $\mathbb{R}$ and A contains each rational number, which of the following must be true?
 - (a) If A is open, then $A = \mathbb{R}$.
 - (b) If A is closed, then $A = \mathbb{R}$.
 - (c) If A is uncountable, then $A = \mathbb{R}$.
 - (d) If A is uncountable, then A is open.
3. Statement 1 | Every permutation is a one-to-one function. Statement 2 | Every function is a permutation.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Find the characteristic of the ring $\mathbb{Z} \times \mathbb{Z}$.
 - (a) 0
 - (b) 3
 - (c) 12
 - (d) 30
5. Find all c in $\mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/(x^3 + x^2 + c)$ is a field.
 - (a) 0
 - (b) 2
 - (c) 1
 - (d) 3
6. The shortest distance from the curve $xy = 8$ to the origin is
 - (a) 4

- (b) 8
(c) 16
(d) $2\sqrt{2}$
7. If f is a linear transformation from the plane to the real numbers and if $f(1, 1) = 1$ and $f(-1, 0) = 2$,
(a) 9
(b) 8
(c) 0
(d) -5
8. Find the characteristic of the ring $\mathbb{Z}_3 \times \mathbb{Z}_3$.
(a) 0
(b) 3
(c) 12
(d) 30
9. Find the generator for the finite field $\mathbb{Z}_7$.
(a) 1
(b) 2
(c) 3
(d) 4

True / False (2 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'The polynomial $x^3 + 2x^2 + 2x + 1$ can be factored into linear factors over $\mathbb{Z}_5$. Find this factorization.' is $(x + 1)(x - 4)(x - 2)$ '.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Simplify

$$\frac{e^x \sin x - \cos x}{e^x \sin x + \cos x}$$

The answer will be a trigonometric function of some simple function of x , like " $\cos 2x$ " or " $\sin(x^3)$ ".

2. What is the smallest possible number of whole 2-by-3 non-overlapping rectangles needed to cover a square region exactly, without extra over-hangs and without gaps?

End of Paper